

As - Home work-

Minimum of all function:-

Book:-

$$f(x, y) = (x + 2y - 7)^2 + (2x + y - 5)^2$$

for $x=1$ and $y=3$ by hit graphi-

here

$$(1 + 6 - 7)^2 + (2 + 3 - 5)^2 = (7 - 7)^2 + (5 - 5)^2 = 0$$

verified!

Kimmelblau:- $f(x, y) = (x^2 + y - 11)^2 + (x + y^2 - 7)^2$

x around 3.85

y around -1.8

$$\left((3.85)^2 + (-1.8) - 11 \right)^2 + \left(3.85 + (-1.8)^2 - 7 \right)^2 = 0.288$$

almost.

second iteration.

for $x = 3.58$

$y = -1.85$

$$\left((3.58)^2 + (-1.85) - 11 \right)^2 + \left(3.58 + (-1.85)^2 - 7 \right)^2 = 0.001$$

↑

global minimum reached!

one more global minimum at $x = -3.77$, $y = -3.279$
 $x = 2.99$, $y = 1.99$

3- Greinank: $f(x, y) = 1 + (x^2 + y^2)/4000 - \cos(x) \cdot \cos(y/\sqrt{2})$

frequency of $\cos(x) \rightarrow 2\pi \approx 6.28$

and $\cos(y/\sqrt{2}) \rightarrow 2\pi/\sqrt{2} = 8.89$

- NO ~~local~~ global minima found

• Acceptance probability $P = e^{-\Delta/T}$
 $\Delta = 0.5$

for $T=1$ $P = e^{(-0.5/1)} \rightarrow 61\%$

$T=10$ $P = e^{(-0.5/10)} \rightarrow 0.95 \rightarrow 95\%$

$T=50$ $P = e^{(-0.5/50)} \rightarrow 0.99 \rightarrow 99\%$

$\uparrow$
 almost all nodes gets accepted.

exponential decay:-

$T=50$, decay = 5% every stage
 stopping at $T=0.001$

$$(0.95)^n \times 0.001/50 = 0.000002$$

$$n = \ln(0.000002) / \ln(0.95) = 211 \text{ stages}$$

$$\text{Total iterations} = 211 \times 100 = \boxed{21100}$$

• Testing Greyscale function -

Best point $x = -3.1090$, $y = -4.477$

Min value = 0.0083